

Venkata Surya Nandan Gudumasu

8341078939 | gvsnandan07@gmail.com | linkedin.com/in/venkata-surya-nandan-g | github.com/SuryaNandan07

EDUCATION

MLR Institute of Technology
B.Tech in Computer Science & Engineering

Hyderabad, India
2023 – 2027

EXPERIENCE

Teaching Intern — Rainbow Primary School, Pragathi Nagar

Hyderabad, India

- Delivered engaging sessions on Block-Based Programming and Introductory Robotics to primary school students.
- Simplified complex programming concepts for young learners, improving student understanding of computational thinking.
- Designed hands-on activities to foster interest in technology and coding among school children.

PROJECTS

Cake Studio

Interactive Design Platform — React.js, Three.js, React Konva

- Developed an interactive cake customization platform with a 2D drag-and-drop designer (React Konva) for placing candles and decorations with real-time updates.
- Built a 3D cake visualization module using Three.js and React Three Fiber with live color/material customization and model rotation controls.

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StreamForge

Full-Stack — React.js, Node.js, Socket.io, FFmpeg, RTMP, HLS

- Built a Twitch-inspired live streaming platform with RTMP-based stream ingestion, FFmpeg transcoding to HLS, and a creator dashboard for managing stream lifecycle.
- Implemented real-time viewer chat using Socket.io and JWT-based authentication, handling the full pipeline from OBS stream ingestion to web video playback.

AI Fitness Planner

AI — React.js, Node.js, MongoDB, Gemini AI, OpenCV

- Built an AI-powered fitness assistant offering personalized workout planning, OpenCV-based posture analysis, and exercise repetition counting, with progress tracked via Recharts dashboards.
- Integrated Gemini AI for personalized recommendations and budget-aware diet suggestions, and designed the user onboarding and goal-management system end-to-end.

Travelio

Full-Stack Web Application | MongoDB, Express, ReactJS, Node.js, n8n

- Developed a travel planning web application to help users discover destinations, plan itineraries, and manage trips.
- Implemented features for destination search, booking management, and trip scheduling using React and Node.js.

BuildForge

Full-Stack Web Application | HTML, CSS, JS, Node.js, Express, MySQL

- Built a web platform connecting users with verified construction professionals (civil, electrical, plumbing), aimed at eliminating scams and improving hiring trust.
- Implemented a three-role system (Admin, Engineer, User) with an admin verification pipeline where engineers submit profiles and are approved before appearing on service pages.

Car Control by Camera

Computer Vision | OpenCV

- Developed a computer vision system using OpenCV to control a car model via real-time camera input.
- Implemented gesture or color detection algorithms to translate visual cues into directional control signals.

CERTIFICATIONS

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| • Data Structures and Algorithms using Java — Infosys Springboard | <i>June 2025</i> |
| • Java (Basic) — HackerRank | <i>Nov 2024</i> |
| • SQL (Basic) — HackerRank | <i>Dec 2024</i> |
| • Problem Solving (Basic) — HackerRank | <i>Jun 2024</i> |

ACHIEVEMENTS

- Operation CodeX — Certificate of Participation for exceptional performance in a coding contest conducted by SQUAD Club at MLR Institute of Technology, Hyderabad (April 2025).

TECHNICAL SKILLS

Languages: Java, Python

Frontend: React.js, Vite, HTML, CSS, JavaScript

Backend: Node.js, Express.js

Databases: MongoDB, MySQL, Supabase

Libraries & Tools: OpenCV, FFmpeg, Blender, Git, GitHub

Concepts: Data Structures & Algorithms, OOP, Computer Vision, AI Integration (Gemini AI), Real-Time Streaming (RTMP/HLS), JWT Authentication